

| 《数据库系统原理》实验报告（1）                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |  |    |  |    |            |
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| 题目：DDL 语言实验                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |  |    |  |    |            |
| 学号                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  | 姓名 |  | 日期 | 2025.04.09 |
| <div>实验环境：<br/>Docker-desktop 4.40.0<br/>oceanbase-ce 4.3.5.1</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |  |    |  |    |            |
| <div>实验步骤及结果截图：<br/>(1)创建实验所需的数据库并使用<br/>create database exp1;<br/>use exp1;<br/><pre>sh-4.4# obclient -uroot@sys -h127.1 -P2881; Welcome to the OceanBase.  Commands end with ; or \g. Your OceanBase connection id is 3221498747 Server version: OceanBase_CE 4.3.5.1 (r101000042025031818-b6d5706eb3d2c5f501c7fa646ddb32f3dc87069) (Built Mar 18 2025 18:13:36)  Copyright (c) 2000, 2018, OceanBase and/or its affiliates. All rights reserved.  Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.  obclient(root@sys)[(none)]&gt; create database exp1; Query OK, 1 row affected (0.033 sec)  obclient(root@sys)[(none)]&gt; use exp1 Database changed obclient(root@sys)[exp1]&gt;</pre></div> <div>(2)表结构设计，根据题目要求创建下面 5 个表（电商系统相关数据表）：<br/>①表 Customers<br/>CREATE TABLE Customers (<br/>CustomerID INT AUTO_INCREMENT NOT NULL,<br/>Name VARCHAR(50) NOT NULL,<br/>Email VARCHAR(100) NOT NULL UNIQUE,<br/>Phone VARCHAR(20) NOT NULL,<br/>RegistrationDate DATE NOT NULL DEFAULT (date(current_timestamp)),<br/>PRIMARY KEY (CustomerID),<br/>-- 检查邮箱是否包含 @ 和 .<br/>CHECK (Email LIKE '%@%.%'),<br/>-- 检查电话（如果非空，长度为 11 且全为数字）<br/>CHECK (LENGTH(Phone) = 11 AND Phone NOT LIKE '%[^0-9]%')<br/>);<br/><pre>obclient(root@sys)[exp1]&gt; CREATE TABLE Customers ( -&gt; CustomerID INT AUTO_INCREMENT NOT NULL, -&gt; Name VARCHAR(50) NOT NULL, -&gt; Email VARCHAR(100) NOT NULL UNIQUE, -&gt; Phone VARCHAR(20) NOT NULL, -&gt; RegistrationDate DATE NOT NULL DEFAULT (date(current_timestamp)), -&gt; PRIMARY KEY (CustomerID), -&gt; -- 检查邮箱是否包含 @ 和 . -&gt; CHECK (Email LIKE '%@%.%'), -&gt; -- 检查电话（如果非空，长度为 11 且全为数字） -&gt; CHECK (LENGTH(Phone) = 11 AND Phone NOT LIKE '%[^0-9]%') -&gt; ); Query OK, 0 rows affected (0.069 sec)</pre></div> <div>②表 Categories<br/>CREATE TABLE Categories (<br/></div> |  |    |  |    |            |

```
CategoryID INT AUTO_INCREMENT NOT NULL,
CategoryName VARCHAR(50) NOT NULL UNIQUE,
PRIMARY KEY (CategoryID)
);
```

```
obclient(root@sys)[exp1]> CREATE TABLE Categories (
-> CategoryID INT AUTO_INCREMENT NOT NULL,
-> CategoryName VARCHAR(50) NOT NULL UNIQUE,
-> PRIMARY KEY (CategoryID)
-> );
Query OK, 0 rows affected (0.061 sec)
```

### ③表 Products

```
CREATE TABLE Products (
ProductID INT AUTO_INCREMENT NOT NULL,
ProductName VARCHAR(100) NOT NULL,
Price DECIMAL(10,2) NOT NULL CHECK (Price > 0),
StockQuantity INT NOT NULL CHECK (StockQuantity >= 0),
CategoryID INT NOT NULL,
PRIMARY KEY (ProductID),
FOREIGN KEY (CategoryID) REFERENCES Categories(CategoryID)
);
```

```
obclient(root@sys)[exp1]> CREATE TABLE Products (
-> ProductID INT AUTO_INCREMENT NOT NULL,
-> ProductName VARCHAR(100) NOT NULL,
-> Price DECIMAL(10,2) NOT NULL CHECK (Price > 0),
-> StockQuantity INT NOT NULL CHECK (StockQuantity >= 0),
-> CategoryID INT NOT NULL,
-> PRIMARY KEY (ProductID),
-> FOREIGN KEY (CategoryID) REFERENCES Categories(CategoryID)
-> );
Query OK, 0 rows affected (0.118 sec)
```

### ④表 Orders

```
CREATE TABLE Orders (
OrderID INT AUTO_INCREMENT NOT NULL,
CustomerID INT NOT NULL,
OrderDate DATETIME NOT NULL DEFAULT current_timestamp,
TotalAmount DECIMAL(10,2) NOT NULL CHECK (TotalAmount >= 0),
PRIMARY KEY (OrderID),
FOREIGN KEY (CustomerID) REFERENCES Customers(CustomerID)
);
```

```
obclient(root@sys)[exp1]> CREATE TABLE Orders (
-> OrderID INT AUTO_INCREMENT NOT NULL,
-> CustomerID INT NOT NULL,
-> OrderDate DATETIME NOT NULL DEFAULT current_timestamp,
-> TotalAmount DECIMAL(10,2) NOT NULL CHECK (TotalAmount >= 0),
-> PRIMARY KEY (OrderID),
-> FOREIGN KEY (CustomerID) REFERENCES Customers(CustomerID)
-> );
Query OK, 0 rows affected (0.159 sec)
```

### ⑤表 OrdersItems

```
CREATE TABLE OrderItems (
OrderItemID INT AUTO_INCREMENT NOT NULL,
```

```

OrderID INT NOT NULL,
ProductID INT NOT NULL,
Quantity INT NOT NULL CHECK (Quantity > 0),
Subtotal DECIMAL(10,2) NOT NULL CHECK (Subtotal > 0),
PRIMARY KEY (OrderItemID),
FOREIGN KEY (OrderID) REFERENCES Orders(OrderID),
FOREIGN KEY (ProductID) REFERENCES Products(ProductID)
);

```

```

obclient(root@sys)[exp1]> CREATE TABLE OrderItems (
->   OrderItemID INT AUTO_INCREMENT NOT NULL,
->   OrderID INT NOT NULL,
->   ProductID INT NOT NULL,
->   Quantity INT NOT NULL CHECK (Quantity > 0),
->   Subtotal DECIMAL(10,2) NOT NULL CHECK (Subtotal > 0),
->   PRIMARY KEY (OrderItemID),
->   FOREIGN KEY (OrderID) REFERENCES Orders(OrderID),
->   FOREIGN KEY (ProductID) REFERENCES Products(ProductID)
-> );
Query OK, 0 rows affected (0.133 sec)

```

触发器：设置  $\text{OrderItems.Subtotal} = \text{Products.Price} * \text{OrderItems.Quantity}$ ，满足字段 Subtotal 的设计要求。

```

DELIMITER //
CREATE TRIGGER before_orderitems_insert
BEFORE INSERT ON OrderItems
FOR EACH ROW
BEGIN
    DECLARE product_price DECIMAL(10,2);
    -- 获取产品价格
    SELECT Price INTO product_price
    FROM Products
    WHERE ProductID = NEW.ProductID;
    -- 设置 Subtotal = Price * Quantity
    SET NEW.Subtotal = product_price * NEW.Quantity;
END //
DELIMITER ;

```

```

obclient(root@sys)[exp1]> -- 触发器：设置OrderItems.Subtotal = Products.Price * OrderItems.Quantity
Query OK, 0 rows affected (0.001 sec)

obclient(root@sys)[exp1]> DELIMITER //
obclient(root@sys)[exp1]> CREATE TRIGGER before_orderitems_insert
-> BEFORE INSERT ON OrderItems
-> FOR EACH ROW
-> BEGIN
->   DECLARE product_price DECIMAL(10,2);
->
->   -- 获取产品价格
->   SELECT Price INTO product_price
->   FROM Products
->   WHERE ProductID = NEW.ProductID;
->
->   -- 设置 Subtotal = Price * Quantity
->   SET NEW.Subtotal = product_price * NEW.Quantity;
-> END //
Query OK, 0 rows affected (0.110 sec)

obclient(root@sys)[exp1]>
obclient(root@sys)[exp1]> DELIMITER ;

```

建立完所有表后进行 `show tables` 展示所有表：

```
obclient(root@sys)[exp1]> show tables;
+-----+
| Tables_in_exp1 |
+-----+
| Categories      |
| Customers       |
| OrderItems      |
| Orders          |
| Products        |
+-----+
5 rows in set (0.008 sec)
```

### (3)表结构查询

desc Customers;

desc Products;

desc Categories;

desc Order;

desc OrderItems;

```
obclient(root@sys)[exp1]> desc Customers;
+-----+-----+-----+-----+-----+-----+
| Field          | Type          | Null | Key | Default        | Extra          |
+-----+-----+-----+-----+-----+-----+
| CustomerID     | int(11)       | NO   | PRI | NULL           | auto_increment |
| Name           | varchar(50)   | NO   |     | NULL           |                |
| Email          | varchar(100)  | NO   | UNI | NULL           |                |
| Phone          | varchar(20)   | NO   |     | NULL           |                |
| RegistrationDate | date          | NO   |     | date(now())    |                |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.022 sec)
```

```
obclient(root@sys)[exp1]> desc Products;
+-----+-----+-----+-----+-----+-----+
| Field          | Type          | Null | Key | Default        | Extra          |
+-----+-----+-----+-----+-----+-----+
| ProductID      | int(11)       | NO   | PRI | NULL           | auto_increment |
| ProductName    | varchar(100)  | NO   |     | NULL           |                |
| Price          | decimal(10,2) | NO   |     | NULL           |                |
| StockQuantity  | int(11)       | NO   |     | NULL           |                |
| CategoryID     | int(11)       | NO   |     | NULL           |                |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.011 sec)
```

```
obclient(root@sys)[exp1]> desc Categories;
+-----+-----+-----+-----+-----+-----+
| Field          | Type          | Null | Key | Default        | Extra          |
+-----+-----+-----+-----+-----+-----+
| CategoryID     | int(11)       | NO   | PRI | NULL           | auto_increment |
| CategoryName   | varchar(50)   | NO   | UNI | NULL           |                |
+-----+-----+-----+-----+-----+-----+
2 rows in set (0.016 sec)
```

```
obclient(root@sys)[exp1]> desc Orders;
+-----+-----+-----+-----+-----+-----+
| Field          | Type          | Null | Key | Default        | Extra          |
+-----+-----+-----+-----+-----+-----+
| OrderID        | int(11)       | NO   | PRI | NULL           | auto_increment |
| CustomerID     | int(11)       | NO   |     | NULL           |                |
| OrderDate      | datetime      | NO   |     | CURRENT_TIMESTAMP |                |
| TotalAmount    | decimal(10,2) | NO   |     | NULL           |                |
+-----+-----+-----+-----+-----+-----+
4 rows in set (0.010 sec)
```

```
obclient(root@sys)[exp1]> desc OrderItems;
```

| Field       | Type          | Null | Key | Default | Extra          |
|-------------|---------------|------|-----|---------|----------------|
| OrderItemID | int(11)       | NO   | PRI | NULL    | auto_increment |
| OrderID     | int(11)       | NO   |     | NULL    |                |
| ProductID   | int(11)       | NO   |     | NULL    |                |
| Quantity    | int(11)       | NO   |     | NULL    |                |
| Subtotal    | decimal(10,2) | NO   |     | NULL    |                |

5 rows in set (0.012 sec)

#### (4)查询建立的约束

```
SELECT table_name, constraint_name, constraint_type
FROM information_schema.TABLE_CONSTRAINTS
WHERE table_name = 'Customers';
```

```
obclient(root@sys)[exp1]> SELECT table_name, constraint_name, constraint_type
-> FROM information_schema.TABLE_CONSTRAINTS
-> WHERE table_name = 'Customers';
```

| table_name | constraint_name                    | constraint_type |
|------------|------------------------------------|-----------------|
| Customers  | PRIMARY                            | PRIMARY KEY     |
| Customers  | Email                              | UNIQUE          |
| Customers  | Customers_OBCHECK_1744206782705287 | CHECK           |
| Customers  | Customers_OBCHECK_1744206782705349 | CHECK           |

4 rows in set (0.081 sec)

```
SELECT table_name, constraint_name, constraint_type
FROM information_schema.TABLE_CONSTRAINTS
WHERE table_name = 'Products';
```

```
obclient(root@sys)[exp1]> SELECT table_name, constraint_name, constraint_type
-> FROM information_schema.TABLE_CONSTRAINTS
-> WHERE table_name = 'Products';
```

| table_name | constraint_name                   | constraint_type |
|------------|-----------------------------------|-----------------|
| Products   | PRIMARY                           | PRIMARY KEY     |
| Products   | Products_OBCHECK_1744208007839641 | CHECK           |
| Products   | Products_OBCHECK_1744208007839735 | CHECK           |
| Products   | Products_OBFK_1744208007840144    | FOREIGN KEY     |

4 rows in set (0.005 sec)

```
SELECT table_name, constraint_name, constraint_type
FROM information_schema.TABLE_CONSTRAINTS
WHERE table_name = 'Categories';
```

```
obclient(root@sys)[exp1]> SELECT table_name, constraint_name, constraint_type
-> FROM information_schema.TABLE_CONSTRAINTS
-> WHERE table_name = 'Categories';
```

| table_name | constraint_name | constraint_type |
|------------|-----------------|-----------------|
| Categories | PRIMARY         | PRIMARY KEY     |
| Categories | CategoryName    | UNIQUE          |

2 rows in set (0.004 sec)

```
SELECT table_name, constraint_name, constraint_type
FROM information_schema.TABLE_CONSTRAINTS
WHERE table_name = 'Orders';
```

```
obclient(root@sys)[exp1]> SELECT table_name, constraint_name, constraint_type
-> FROM information_schema.TABLE_CONSTRAINTS
-> WHERE table_name = 'Orders';
```

| table_name | constraint_name                 | constraint_type |
|------------|---------------------------------|-----------------|
| Orders     | PRIMARY                         | PRIMARY KEY     |
| Orders     | Orders_OBCHECK_1744213010874996 | CHECK           |
| Orders     | Orders_OBFK_1744213010875157    | FOREIGN KEY     |

3 rows in set (0.003 sec)

```
SELECT table_name, constraint_name, constraint_type
FROM information_schema.TABLE_CONSTRAINTS
WHERE table_name = 'OrderItems';
```

```
obclient(root@sys)[exp1]> SELECT table_name, constraint_name, constraint_type
-> FROM information_schema.TABLE_CONSTRAINTS
-> WHERE table_name = 'OrderItems';
```

| table_name | constraint_name                     | constraint_type |
|------------|-------------------------------------|-----------------|
| OrderItems | PRIMARY                             | PRIMARY KEY     |
| OrderItems | OrderItems_OBCHECK_1744214062283249 | CHECK           |
| OrderItems | OrderItems_OBCHECK_1744214062283507 | CHECK           |
| OrderItems | OrderItems_OBFK_1744214062292714    | FOREIGN KEY     |
| OrderItems | OrderItems_OBFK_1744214062283745    | FOREIGN KEY     |

5 rows in set (0.004 sec)

##### (5)表结构修改与操作

①在 **Customers** 表中添加一个新字段 **LastLoginDate**，类型为 **DATETIME**，当客户每次登录时，该字段自动更新为当前时间。需创建触发器实现此功能。

第一步先进行表的修改，添加一个新字段：

```
ALTER TABLE Customers
ADD LastLoginDate DATETIME NOT NULL;
```

```
obclient(root@sys)[exp1]> ALTER TABLE Customers
-> ADD LastLoginDate DATETIME NOT NULL;
Query OK, 0 rows affected (0.118 sec)

obclient(root@sys)[exp1]> desc Customers;
```

| Field            | Type         | Null | Key | Default     | Extra          |
|------------------|--------------|------|-----|-------------|----------------|
| CustomerID       | int(11)      | NO   | PRI | NULL        | auto_increment |
| Name             | char(15)     | NO   |     | NULL        |                |
| Email            | varchar(100) | NO   | UNI | NULL        |                |
| Phone            | varchar(20)  | NO   |     | NULL        |                |
| RegistrationDate | date         | NO   |     | date(now()) |                |
| LastLoginDate    | datetime     | NO   |     | NULL        |                |

6 rows in set (0.022 sec)

第一个触发器会在向 **Customers** 表插入新记录（比如新用户注册）之前被触发。在插入新记录时，自动将新记录的 **LastLoginDate** 字段设置为当前时间：

```
DELIMITER //
CREATE TRIGGER before_customers_insert
BEFORE INSERT ON Customers
FOR EACH ROW
```

```
BEGIN
-- 新用户注册时，设置 LastLoginDate 为当前时间
SET NEW.LastLoginDate = CURRENT_TIMESTAMP;
END //
DELIMITER ;
```

第二个触发器会在 Customers 表的记录被更新之前触发。在更新记录，即用户登录时，自动将 LastLoginDate 字段设置为当前时间：

```
DELIMITER //
CREATE TRIGGER before_customers_update
BEFORE UPDATE ON Customers
FOR EACH ROW
BEGIN
-- 在用户登录（更新）时，自动设置 LastLoginDate 为当前时间
SET NEW.LastLoginDate = current_timestamp;
END //
DELIMITER ;
```

```
obclient(root@sys)[exp1]> DELIMITER //
obclient(root@sys)[exp1]> CREATE TRIGGER before_customers_insert
-> BEFORE INSERT ON Customers
-> FOR EACH ROW
-> BEGIN
-> -- 新用户注册时，设置 LastLoginDate 为当前时间
-> SET NEW.LastLoginDate = CURRENT_TIMESTAMP;
-> END //
Query OK, 0 rows affected (0.120 sec)

obclient(root@sys)[exp1]> DELIMITER ;
obclient(root@sys)[exp1]> DELIMITER //
obclient(root@sys)[exp1]> CREATE TRIGGER before_customers_update
-> BEFORE UPDATE ON Customers
-> FOR EACH ROW
-> BEGIN
-> -- 在用户登录（更新）时，自动设置 LastLoginDate 为当前时间
-> SET NEW.LastLoginDate = current_timestamp;
-> END //
Query OK, 0 rows affected (0.129 sec)

obclient(root@sys)[exp1]> DELIMITER ;
```

## ②将 Customers 中的 name 由 varchar 改为 char(15)。

对于 Name 列的值，如果它的长度（字符数）超过 15 个字符，就只保留前 15 个字符，为更改其数据类型做准备：

```
UPDATE Customers
SET Name = LEFT(Name, 15)
WHERE LENGTH(Name) > 15;
然后将 Name 列的数据类型改为 CHAR(15):
ALTER TABLE Customers
MODIFY COLUMN Name CHAR(15) NOT NULL;
```

```
obclient(root@sys)[exp1]> UPDATE Customers
-> SET Name = LEFT(Name, 15)
-> WHERE LENGTH(Name) > 15;
Query OK, 0 rows affected (0.028 sec)
Rows matched: 0 Changed: 0 Warnings: 0

obclient(root@sys)[exp1]> ALTER TABLE Customers
-> MODIFY COLUMN Name CHAR(15) NOT NULL;
Query OK, 0 rows affected (1.189 sec)

obclient(root@sys)[exp1]> desc Customers;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| CustomerID | int(11) | NO | PRI | NULL | auto_increment |
| Name | char(15) | NO | | NULL | |
| Email | varchar(100) | NO | UNI | NULL | |
| Phone | varchar(20) | NO | | NULL | |
| RegistrationDate | date | NO | | date(now()) | |
| LastLoginDate | datetime | NO | | NULL | |
+-----+-----+-----+-----+-----+-----+
6 rows in set (0.028 sec)
```

③修改 Products 表，添加一个 IsFeatured 字段，类型为 BOOLEAN，默认值为 FALSE。同时，创建一个索引，用于快速查询特色商品（IsFeatured 为 TRUE 的商品）。

向 Products 表添加一个新字段 IsFeatured，数据类型为 BOOLEAN：

ALTER TABLE Products

ADD IsFeatured BOOLEAN NOT NULL DEFAULT FALSE;

```
obclient(root@sys)[exp1]> ALTER TABLE Products
-> ADD IsFeatured BOOLEAN NOT NULL DEFAULT FALSE;
Query OK, 0 rows affected (0.117 sec)

obclient(root@sys)[exp1]> desc Products;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| ProductID | int(11) | NO | PRI | NULL | auto_increment |
| ProductName | varchar(100) | NO | | NULL | |
| Price | decimal(10,2) | NO | | NULL | |
| StockQuantity | int(11) | NO | | NULL | |
| CategoryID | int(11) | NO | | NULL | |
| IsFeatured | tinyint(1) | NO | | 0 | |
+-----+-----+-----+-----+-----+-----+
6 rows in set (0.003 sec)
```

在 Products 表的 IsFeatured 列上创建一个索引，找到表中 IsFeatured 列为 TRUE 所有数据，并显示该索引：

CREATE INDEX idx\_isfeatured ON Products(IsFeatured);

SHOW INDEX FROM Products;

```
obclient(root@sys)[exp1]> CREATE INDEX idx_isfeatured ON Products(IsFeatured);
Query OK, 0 rows affected (0.324 sec)

obclient(root@sys)[exp1]> SHOW INDEX FROM Products;
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| Table | Non_unique | Key_name | Seq_in_index | Column_name | Collation | Cardinality | Sub_part | Packed | Null | Index_type | Comment | Index_comment | Visible | Expression |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| Products | 0 | PRIMARY | 1 | ProductID | A | NULL | NULL | NULL | NULL | BTREE | available | | YES | NULL |
| Products | 1 | idx_isfeatured | 1 | IsFeatured | A | NULL | NULL | NULL | NULL | BTREE | available | | YES | NULL |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
12 rows in set (0.022 sec)
```

④删除 Categories 表中所有没有关联商品的分类记录，同时要保证删除操作符合数据库的参照完整性。需创建一个存储过程实现此功能。

存储过程 DeleteOrphanedCategories 的目的是清理 Categories 表中没有关联商品的记录。步骤是：定义变量 deleted\_rows 记录删除行数；设置错误处理器，确保异常时回滚事务并抛出错误；开启事务，保证操作的原子性；使用 NOT EXISTS 子查询找出没有商品的分类并删除；记录删除的行数；提交事务，确认更改；返回删除的行数信息。

DELIMITER //



```

CREATE PROCEDURE DeleteOrphanedCategories()
BEGIN
    -- 声明变量用于错误处理和记录删除行数
    DECLARE deleted_rows INT DEFAULT 0;
    DECLARE EXIT HANDLER FOR SQLEXCEPTION
    BEGIN
        -- 错误发生时回滚事务并返回错误信息
        ROLLBACK;
        SIGNAL SQLSTATE '45000'
        SET MESSAGE_TEXT = 'An error occurred while deleting orphaned categories';
    END;
    -- 开启事务，确保操作原子性
    START TRANSACTION;
    -- 删除没有关联商品的分类
    DELETE FROM Categories
    WHERE NOT EXISTS (
        SELECT 1
        FROM Products
        WHERE Products.CategoryID = Categories.CategoryID
    );
    -- 获取删除的行数
    SET deleted_rows = ROW_COUNT();
    -- 提交事务
    COMMIT;
    -- 返回删除的行数，用于验证
    SELECT CONCAT('Deleted ', deleted_rows, ' orphaned categories') AS Result;
END //
DELIMITER ;

```

```

obclient(root@sys)[exp1]> DELIMITER //
obclient(root@sys)[exp1]> CREATE PROCEDURE DeleteOrphanedCategories()
-> BEGIN
->     -- 声明变量用于错误处理和记录删除行数
->     DECLARE deleted_rows INT DEFAULT 0;
->     DECLARE EXIT HANDLER FOR SQLEXCEPTION
->     BEGIN
->         -- 错误发生时回滚事务并返回错误信息
->         ROLLBACK;
->         SIGNAL SQLSTATE '45000'
->         SET MESSAGE_TEXT = 'An error occurred while deleting orphaned categories';
->     END;
->
->     -- 开启事务，确保操作原子性
->     START TRANSACTION;
->
->     -- 删除没有关联商品的分类
->     DELETE FROM Categories
->     WHERE NOT EXISTS (
->         SELECT 1
->         FROM Products
->         WHERE Products.CategoryID = Categories.CategoryID
->     );
->
->     -- 获取删除的行数
->     SET deleted_rows = ROW_COUNT();
->
->     -- 提交事务
->     COMMIT;
->
->     -- 返回删除的行数，用于验证
->     SELECT CONCAT('Deleted ', deleted_rows, ' orphaned categories') AS Result;
-> END //
Query OK, 0 rows affected (0.119 sec)

obclient(root@sys)[exp1]> DELIMITER ;

```

执行刚刚定义的存储过程，触发删除操作并返回结果：

CALL DeleteOrphanedCategories();

```
obclient(root@sys)[exp1]> CALL DeleteOrphanedCategories();
+-----+
| Result |
+-----+
| Deleted 0 orphaned categories |
+-----+
1 row in set (0.178 sec)

Query OK, 0 rows affected (0.178 sec)
```

出现的问题：

#### ①DATE 型变量设置默认值会报错

错误发生在 RegistrationDate DATE NOT NULL DEFAULT (date(current\_timestamp))。该版本 oceanbase-ce 不支持直接在 DEFAULT 中使用 date(current\_timestamp) 这样的默认值表达式。

```
obclient(root@sys)[exp1]> CREATE TABLE Customers (
--> CustomerID INT AUTO_INCREMENT PRIMARY KEY,
--> Name VARCHAR(50) NOT NULL,
--> Email VARCHAR(100) NOT NULL UNIQUE,
--> Phone VARCHAR(20) NULL,
--> RegistrationDate DATE NOT NULL DEFAULT (date(current_timestamp)),
--> -- 检查邮箱是否包含 @ 和 .
--> CHECK (Email LIKE '%@%.%'),
--> -- 检查电话 (如果非空, 长度为 11 且全为数字)
--> CHECK (Phone IS NULL OR (LENGTH(Phone) = 11 AND Phone NOT LIKE '%[^0-9]%')),
--> );
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your OceanBase version for the right syntax to use near '(date(current_timestamp)),
--  ???????? @ ? .
CHECK (En' at line 6
[172.17.0.2:2882] [2025-04-09 13:34:30.051382] [YB42AC110002-0006325861968D05-0-0]
```

#### ②列定义发生语法错误

错误发生在：

```
CREATE TABLE Categories (
    CategoryID INT AUTO_INCREMENT NOT NULL,
    CategoryName VARCHAR(50) NOT NULL UNIQUE,
    PRIMARY KEY (CategoryID),
);
```

试图创建 Categories 表，但由于语法错误导致失败。

```
obclient(root@sys)[exp1]> CREATE TABLE Categories (
--> CategoryID INT AUTO_INCREMENT NOT NULL,
--> CategoryName VARCHAR(50) NOT NULL UNIQUE,
--> PRIMARY KEY (CategoryID),
--> );
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your OceanBase version for the right syntax to use near ')' at line 5
[172.17.0.2:2882] [2025-04-09 13:58:22.136977] [YB42AC110002-0006325896280601-0-0]
```

#### ③查询语句约束信息错误

SQL 语句试图从 information\_schema.TABLE\_CONSTRAINTS 表中查询 Customers 表的约束信息，但数据库返回了错误信息，错误提示表明，在 WHERE 子句中，Customers 被当作列名处理，但实际上它是一个表名，导致查询失败。

```
obclient(root@sys)[exp1]> SELECT table_name, constraint_name, constraint_type
--> FROM information_schema.TABLE_CONSTRAINTS
--> WHERE table_name = Customers;
ERROR 1054 (42S22): Unknown column 'Customers' in 'where clause'
[172.17.0.2:2882] [2025-04-09 16:24:29.901536] [YB42AC110002-0006325898580603-0-0]
```

#### ④创建索引失败

SQL 语句试图在 Products 表上创建名为 idx\_isfeatured 的索引，但数据库返回了错误信息，错误发生在第 1 行，提示语法问题，具体是 Products 的使用不正确。

```
obclient(root@sys)[exp1]> CREATE INDEX idx_isfeatured ON Products;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your OceanBase version for the right syntax to use near 'Products' at line 1
[172.17.0.2:2882] [2025-04-10 10:08:05.096276] [YB42AC110002-000632698BFDE5A9-0-0]
```

**解决方案：**

**①DATE 型变量设置默认值会报错问题的解决**

按照要求对 oceanbase-ce 进行升级，升到 4.3.5.1 后成功运行。

**②列定义发生语法错误问题的解决**

经检查发现，“PRIMARY KEY (CategoryID),”后面的逗号是多余的，正确的语句应该直接以)结束表定义。在 CREATE TABLE 语句中，列定义或约束列表的最后一个元素后不能有逗号。删去逗号后就可以成功创建。

**③查询语句约束信息错误问题的解决**

WHERE 子句试图筛选表名为 Customers 的记录。问题在于 Customers 没有用单引号或双引号括起来。在 SQL 中，字符串值（包括表名）需要用引号括起来，否则数据库会将其视为列名或标识符。这里，Customers 被错误地解析为列名，而 TABLE\_CONSTRAINTS 表中没有名为 Customers 的列，因此报错。将 Customers 加上引号后结果就正确了。

**④创建索引失败问题的解决**

具体问题是语句缺少索引所基于的列名。在 SQL 中，CREATE INDEX 语句必须明确指定索引要应用于表的哪些列，例如 ON Products(IsFeatured)，改成这样后就可以成功创建。